# Reinstatement out of equilibrium: shock speed and the destination of new work

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**Abstract**

A previous paper by the author closes a spatial task-based model for technological impact on labor as a static equilibrium and reads its consequences as comparative statics between two stationary points. This paper turns the same model into laws of motion and asks what the static account cannot: how the transition unfolds in time, and how its outcome depends on the speed of the shock. The technology matures gradually, as a logistic diffusion over a calendar window, so displacement and the new tasks it seeds on the boundary of the technology field arrive as a flow rather than at once. The dynamics require one new object, a stock of *unbound* task mass. It holds the seeded work that has been created but not yet attached to any occupation, the third fate of new tasks that the static model creates and discards at every evaluation. Its trajectory $U(t)$ is the transitional mismatch between destruction and creation, and it is the model's central dynamic observable. Binding draws the stock down. New work seeks the occupations whose held capabilities best match it, but each occupation absorbs only at a rate limited by its size, so a saturated best match leaves a residue that cascades to the next-best match. A single ratio governs the outcome, the tempo of the shock against the tempo of redistribution. When redistribution keeps pace, the high-skill specialists whose capabilities fit the frontier bind the reinstated work. When the shock outruns redistribution, the small best-matched occupations fall behind and the large, lower-skilled occupations take the mass instead, a forced downgrading. The same technology and the same field thus yield materially different distributions of work depending only on the tempo, and the final allocations of the slow and fast regimes correlate at 0.37. The two regimes are bracketed by the proved limits of the binding law, and the destination moves monotonically between the limits with the tempo. This path-dependence of the destination is invisible to the equilibrium framework, whose fixed point sees only the endpoint. The realisation is illustrative and inherits the calibration of Storck (2026).

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# 1 Introduction

A technological breakthrough can happen overnight. Its consequences take years to unfold, and while they do, the labour market is already responding. Firms invest, new work practices spread, training programmes equip engineers and workers with the skills the technology demands, and schools carry the new knowledge to the next cohort. As the technology diffuses and matures it displaces old tasks and introduces new ones. Some workers find new work where they stand, others move to reach it. These processes run in parallel, and each runs at its own speed.

Research has measured these speeds separately. The diffusion literature has timed the technology since Griliches (1957) fitted logistic curves to the spread of hybrid corn. New technologies take decades to reach their ceilings, general-purpose technologies deliver their gains a generation after they arrive, and adoption lags differ widely across countries (David 1990; Comin and Hobijn 2010). The mobility literature has timed the workers. Occupational moves are slow and costly because much of human capital is specific to tasks, and a mover loses the part that does not transfer (Kambourov and Manovskii 2009; Gathmann and Schönberg 2010). The task-based tradition supplies the content of the shock. Automation displaces labour from existing tasks, the creation of new tasks reinstates it, and the balance between the two sets the demand for labour (Acemoglu and Restrepo 2018; Acemoglu and Restrepo 2019). Each clock has been measured on its own. No framework runs them against each other and asks who ends up with the new work.

The task-based framework reads the consequences of a technology as the difference between two rest points. Its dynamics, where it has them, belong to the technology, to which tasks are automated and when new ones arrive, while workers reach their new allocation instantaneously (Acemoglu and Restrepo 2019). The intermediate path of the allocation drops out, and with it the cost of the transition and any dependence of the outcome on the speed of the shock. Yet the transition is where the human consequences of automation are felt. The central claim of this paper is that the speed of the shock relative to the speed of adjustment determines not merely how costly the transition is but *who ends up doing the reinstated work.*

The framework we build on is developed in two papers. Storck and Andersson (2026) map tasks and occupations from O*NET statements onto a two-dimensional disk, where each occupation is a bundle of tasks. Storck (2026) builds the model of technological change on that geometry. The market price of skill varies over the space as a directional field, and a technology arrives as a Gaussian augmentation field. Automation removes the high-priced tasks of a bundle. Reinstatement seeds new tasks on the gradient boundary of the field, and they survive as human work where labour remains the cheaper producer. That model is closed as a static equilibrium, with the adjustment dynamics left explicitly for later work. The rise-and-fall of employment which Bessen (2019) documents over time appears in it only as a cross-section over the demand elasticity, not as a trajectory. This paper is that later work.

We make one structural addition and otherwise reuse the static model unchanged. That model creates new task mass on the boundary ring and lets it meet three fates. Capital captures one part and occupations bind another, while the unbound remainder is created and discarded at every evaluation, with no home in the model. The dynamic model gives it a home, a stock $U(\mathbf{r}, t)$ of unbound task mass that fills as the technology matures and drains as occupations bind it. This stock is the only genuinely new object; population, price, displacement, seeding, and the bundle operator are the static model's, now read as rates rather than as a solved equilibrium. The trajectory of the stock, $U(t)$, is the transitional mismatch between destruction and creation, and it is the first observable the static model cannot produce.

The second, and the paper's central result, concerns binding. Unbound work seeks the occupations whose held capabilities best match it, but an occupation can take on new work only at a rate limited by its size, so a small best match saturates and leaves a residue for the next-best match. The shock seeds new work at one tempo and occupations bind it at another, so the realised allocation depends on a race. We show that the ratio of the two tempos governs not only the height and duration of the transitional mismatch $U(t)$ but the final destination of the work. A gradual shock lets the best-matched occupations keep pace, and the high-skill specialists whose capabilities fit the frontier bind the reinstated work. A shock that outruns redistribution forces the work onto the large, lower-skilled occupations that can absorb it fastest, a downgrading that the gradual transition avoids. The two regimes run through the same technology and the same field, and their final allocations of reinstated work correlate at 0.37. Both endpoints are proved as limits of the binding law. Absorption faster than the shock converges to the match-weighted allocation and absorption slower than the shock to initial size shares, and the realised destination moves monotonically between the two as the tempo varies (Section 4.3). The destination itself is path-dependent, and the equilibrium framework sees none of it.

That reallocation takes time and shapes equilibrium is an old observation. In the islands tradition workers move slowly between markets and unemployment is the reallocation in progress (Lucas and Prescott 1974; Alvarez and Shimer 2011); distance appears as a cost, but there is no space in which the shock and the workers' capabilities are located together. Şahin et al. (2014) measure the unemployment produced when job seekers search in the wrong markets: workers waiting for work. Guvenen et al. (2020) measure mismatch inside the match, as the distance between the skills an occupation requires and those its worker brings. The unbound stock of this paper is the reverse object, work that has been created but not yet attached to any occupation, mismatch upstream of any vacancy. Its hump is the model's analogue of their cyclical index, driven by task destruction and creation rather than by the business cycle. Closest on the question of tempo, Adão, Beraja, and

Pandalai-Nayar (2024) relate the speed of a technological transition to skill specificity. When the benefited activities require skills far from the rest of the economy, incumbents cannot reallocate and demand waits for the entry of new generations. In their model the economy converges to the endpoint the technology pins down, regardless of the path. Here the tempo of the shock races a fixed speed of reallocation, and the endpoint moves with it. The same technology, run slowly or quickly through the same economy, binds the reinstated work to different occupations. The tempo is also a policy margin. Guerreiro, Rebelo, and Teles (2022) find it optimal to tax robots while the workers who cannot move remain in the labour force, and Beraja and Zorzi (2025) to slow automation while displaced workers reallocate under borrowing constraints. Our result gives that lever a second consequence: pacing moves the destination of the work. The calibration remains illustrative in the discipline of Storck (2026), and we make no causal claim from it.

# 2 The static foundation

We recall only what the dynamics require. Storck and Andersson (2026) construct and validate the geometry, the disk, its coordinates, and the mapping of tasks and occupations into it. Storck (2026) builds the model of technological change on that geometry and supplies the calibration used here.

Tasks lie on the unit disk in polar coordinates $\mathbf{r} = (\xi, \chi)$, with $\xi$ a domain orientation and $\chi$ a depth. The market price of skill is a directional field

$$\ln \Pi(\mathbf{r}) = m_0 + m_1 \cos\xi + m_2 \sin\xi + \chi\,(m_3 + m_4 \cos\xi + m_5 \sin\xi), \tag{1}$$

high in the socio-cognitive directions and low in the service and manual ones (Storck and Andersson 2026). The pattern is the observed wage structure read spatially. Estimated on the occupational wage cross-section ($N = 785$; HC3 standard errors in parentheses), the field is

$$\ln \Pi(\mathbf{r}) = \underset{(0.036)}{3.420} + \underset{(0.048)}{0.192}\,\cos\xi + \underset{(0.051)}{0.089}\,\sin\xi + \chi\Big(\underset{(0.090)}{-0.065} + \underset{(0.112)}{0.008}\,\cos\xi + \underset{(0.134)}{0.891}\,\sin\xi\Big), \tag{2}$$

with $R^2 = 0.523$; the well-paid professional and managerial occupations sit in the socio-cognitive directions of the disk. Two structural checks discipline the form. The isotropic depth term is indistinguishable from zero ($m_3 = -0.065$, $p = 0.47$), so depth pays only through direction, and a level-balanced second-harmonic test does not reject a single first harmonic for the depth return ($p = 0.44$), which peaks at 89.5° and is positive only in the northern half-plane. Appendix A carries the estimation detail and the provenance of every inherited parameter. Depth is what Adão, Beraja, and Pandalai-Nayar (2024) call skill specificity. The geometry gives it a direction, and the price field decides where specificity pays. A technology is a Gaussian field $\phi_K(\mathbf{r}) = A_K \exp(-\frac{1}{2}\|\mathbf{r} - \mathbf{p}_K\|^2/z_K^2)$, with maturity $A_K$, centre $\mathbf{p}_K$, and reach $z_K$. Capital bears a task only where it is the cheaper producer, so the operated share is

$$a(\mathbf{r}) = \Big[1 + \exp\Big(-\Big(s_K\phi_K(\mathbf{r}) - R/\Pi(\mathbf{r})\Big)/\tau\Big)\Big]^{-1}, \tag{3}$$

where $\tau$ is the width of the adoption margin. The condition crosses first where the price $\Pi$ is high, so automation takes the dearest work. Each occupation $o$ is a bundle $b_o(\mathbf{r})$ with centroid $\mu_o$; its displaced mass is $D_o = \int b_o a$, and the aggregate displacement is $\Delta\Gamma^D = \sum_o L_o D_o$.

Reinstatement creates new task mass $M = \gamma\, \Delta\Gamma^D$ on the gradient ring of the field, $\hat{g}(\mathbf{r}) = \|\nabla\phi_K\|/\int\|\nabla\phi_K\|$, which peaks at distance $z_K$ from the centre. A seeded task survives as human work only where the operated share is low, through the survival gate $(1-a)$: deep in the core, where capital dominates, the same seed is captured by capital. The surviving seed density is therefore $\hat{g}\,(1-a)$, concentrated on the boundary crescent. Figure 1 draws this inherited economy from the pipeline: the price contours, the occupation centroids, and the technology field whose gradient ring carries the seeding.

This construction generalises the reinstatement of Acemoglu and Restrepo (2018), and the geometry of their model motivates ours. Production there combines a unit measure of tasks on an interval. Automation raises a lower threshold below which tasks can be produced by capital, and new tasks arrive at the upper endpoint, so reinstatement is boundary creation already on the line. Whether an automatable task is actually produced by capital is a cost comparison, capital where the rental rate beats the effective wage, and equation (3) makes the same comparison pointwise on the disk. Two features of their boundary are assumptions that the disk turns into outcomes. Labour's comparative advantage grows with the task index, so the new task at the endpoint is the one where labour's advantage is greatest, and a bound on the capital stock guarantees that labour adopts it immediately (Acemoglu and Restrepo 2018). Here the endpoint becomes the gradient ring, the ordering becomes the price field, and the guarantee becomes the gate, which prices survival region by region. The new tasks in which capital holds the advantage, noted and set aside in Acemoglu and Restrepo (2019), are the captured fate $s \cdot a$. The generalisation is what opens the destination question. On the line new work has one place to appear. On the ring it appears in every direction around the field, and which occupations end up holding it becomes an economic outcome rather than a location fixed by construction.

An occupation can take a seeded task only where its capabilities match the task's requirements. The implemented readiness is symmetric and local,

$$e_o(\mathbf{r}) = \exp\Big[-\Big(\delta_o(\mathbf{r}) + \lambda_{\text{over}}\,\bar{\delta}_o(\mathbf{r})\Big)/\ell\Big]\, \exp\Big(-\|\mathbf{r} - \mu_o\|/\rho\Big), \tag{4}$$

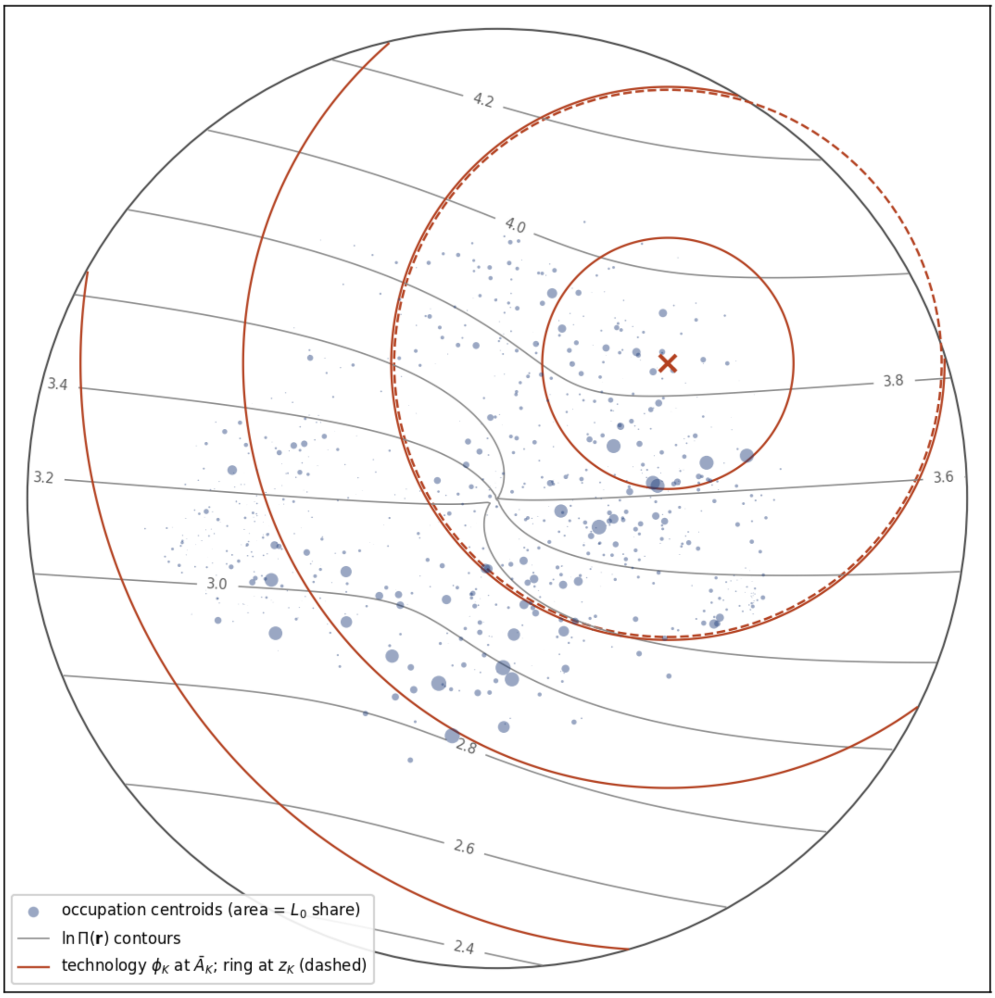


Figure 1: The inherited economy. Grey: contours of the estimated log price field, eq. (2). Blue: occupation centroids, area proportional to pre-shock employment share $L_0$. Red: the calibrated technology field $\phi_K$ at mature amplitude, its centre marked and the gradient ring at radius $z_K$ dashed; the surviving seed concentrates where the ring meets the low-priced side of the adoption boundary. Generated from the same objects the dynamic scripts consume.

where $\delta_o = \sum_k v_k \max(q_k(\mathbf{r}) - q_{o,k}, 0)$ is the $v$-weighted capability deficit, $\bar{\delta}_o$ the corresponding surplus, $\ell$ the acquisition scale, $\rho$ the attachment locality, and $\lambda_{\text{over}}$ the over-qualification weight. Each parameter has a stated derivation. The requirement fields $q_k$ are capability planes over the disk, restated in Appendix A. The weights $v = (0.328, 0.049)$ on the priced clusters $K = \{S1, S2\}$ are wage returns from a replication of the mediation regression of Storck and Andersson (2026); unpriced or negative components are not carried. The scale $\ell = 0.133$ is one within-direction standard deviation of the priced-capability index, the deficit at which readiness falls to $e^{-1}$. The attachment capacity of a location is $C(\mathbf{r}) = \sum_o L_o e_o(\mathbf{r})$. This primitive is shared verbatim between the two papers and between the static and dynamic layers of the implementation.

All parameters of this foundation are inherited unchanged from the economy table of Storck (2026): $R = 18$, $\tau = 0.08$, $\beta = 0.5$ (congestion), $\gamma = 0.5$, $\ell = 0.133$, $\rho = 0.5$, $\lambda_{\text{over}} = 1$. The mobility scale $\nu$ (one standard deviation of baseline occupation value per logit unit) and the move cost $c_m$ (the median move costs one $\nu$) follow the same rules, evaluated here at the pre-shock baseline $A_K = 0$, the state the dynamics start from, giving $\nu = 11.8$ and $c_m = 23.0$. That table evaluates the same rules at the calibrated technology and reports 11.6 and 22.6. The two per cent difference moves no result at its reported precision. Table 2 in Appendix A consolidates every parameter of the economy with its provenance: estimated, calibrated, rule-derived, normalised, or swept.

Two features of this foundation are load-bearing for what follows. First, displacement falls *within* occupations, on the interior margin $a(\mathbf{r})$, not between regions. Second, the survival gate places the surviving reinstatement on the low-priced side of the boundary. Humans retain the work that is not worth automating, which in this price field is the cheaper, lower-$\Pi$ work. The destination results below interact with this second feature, and we return to it in the discussion.

# 3 From equilibrium to motion

The static model computes a rest point $L = F(L)$. The present model attaches calendar time to the same primitives and studies the path of the allocation while technology matures. Technology maturity, population, unbound task mass, and occupational task content become state variables. Displacement, seeding, binding, and reallocation become transition rates. The task geometry supplies the constitutive laws for these rates. Prices set adoption, the boundary field locates reinstatement, readiness ranks viable destinations, and occupation size limits absorption. The resulting state accounting gives the transition an observable stock, $U(\mathbf{r}, t)$, which records new human work before any occupation has taken it on.

## 3.1 Transition accounting and the unbound stock

In the static model the seeded mass $M$ meets three fates. Capital captures a part $s \cdot a$, occupations bind a part $s \cdot (1 - a)\,\Phi$, and a part $s \cdot (1 - a)\,(1 - \Phi)$ remains unbound, where $s = M\hat{g}$ is the seed density and $\Phi$ is the bound share; the three parts sum to $s$. The unbound part performs no work and does not enter the density; it is created and discarded at each evaluation. The dynamic model gives it a stock,

$$\partial_t U(\mathbf{r}, t) = \dot{s}(\mathbf{r}, t) - \sum_o \iota_o(\mathbf{r}, t), \qquad \dot{s}(\mathbf{r}, t) = \gamma\,[\partial_t\,\Delta\Gamma^D]_+\,\hat{g}(\mathbf{r})\,(1 - a(\mathbf{r}, t)), \tag{5}$$

where $\dot{s}$ is the seeding rate and $\iota_o$ the binding flow to occupation $o$ (§3.3). The seeding follows the *rate* of displacement. New tasks enter while the technology matures. Seeding

$$\begin{tikzcd}
\text{Technology maturation } A_K(t) \arrow[r, "\partial_t \Delta\Gamma^D"] & \text{Surviving seed flow } \dot{s}(\mathbf{r},t) \arrow[d] \\
& \boxed{\text{Unbound stock } U(\mathbf{r},t)} \arrow[d, "\iota_o(\mathbf{r},t)"] \\
& \text{Occupation-bound task mass } r_o(t) \arrow[d, phantom, "\circlearrowleft"] \\
& \text{values } W_o(t) \text{ and population } L_o(t)
\end{tikzcd}$$

Figure 2: Transition accounting for displaced and reinstated work. The stock $U(\mathbf{r},t)$ carries surviving new task mass until occupations bind it through the flows $\iota_o(\mathbf{r},t)$. Bound mass changes occupational values and population adjustment.

ceases once displacement plateaus. The unbound stock is the sole structural addition; the remaining objects are the static model's primitives read as rates.

Figure 2 gives the same transition accounting as a stock and flow diagram. Automation creates a timed inflow of surviving new task mass. Binding flows draw from $U(\mathbf{r},t)$ and add occupation-bound task mass. The accumulation of bound mass changes occupational values and subsequent adjustment incentives.

Task mass enters through seeding and leaves $U$ through binding. Population is conserved, $\sum_o \dot{L}_o = 0$, and the created task mass is assigned among three fates: capital capture, occupation binding, and unbound carry-over. The stock $U$ closes this ledger by giving the carry-over term a state variable.

## 3.2 Population and maturation

The population moves toward the static reallocation target, but from the *current* distribution rather than a frozen baseline:

$$\dot{L}_o = \tfrac{1}{\theta_L}\Big(T_o(L,t) - L_o\Big), \qquad T_o(L,t) = \sum_{o'} L_{o'}(t)\, P\Big(o \mid o';\, W(L,t)\Big), \tag{6}$$

with $P$ a row-stochastic origin–destination logit, so $\sum_o \dot{L}_o = 0$ exactly. The timescale $\theta_L$ is a calibrated speed in interpretable units. It replaces the static solver's damping and inherits nothing from it. First order is deliberate because it yields lags and gaps but no overshoot.

The technology matures as a logistic diffusion anchored in calendar time, the S-curve that has described technology adoption since Griliches (1957). With $t$ in years and a shock window $T_{\text{shock}}$, the slope is set so that $A_K$ rises from five to ninety-five per cent of its mature value over $T_{\text{shock}}$ years, centred at $T_{\text{shock}}/2$:

$$A_K(t) = \frac{\bar{A}_K}{1 + \exp\Big( - (5.88/T_{\text{shock}})\,(t - T_{\text{shock}}/2)\Big)}. \tag{7}$$

Anchoring the shock in calendar time makes $\theta_L$ and the binding timescale below calendar quantities, so that the outcome is governed by the *ratio* of the shock tempo $T_{\text{shock}}$ to the redistribution tempo. We set $T_{\text{shock}} = 5$ years throughout. The value is a normalisation of the time axis, not a calibration: every rate in the model scales as $1/\theta_L$ or $1/\theta_{\text{abs}}$ or is set by the window, so only the ratio enters the results, and the sweep over the redistribution tempo at fixed $T_{\text{shock}}$ traverses it.

## 3.3 Binding: match allocates, size limits the rate

The binding flow $\iota_o$ rests on a separation that proved load-bearing. Match sets *where* the unbound mass goes, and size sets *how fast* it binds. At each location the mass is attracted to the best match through a sharpened share, and each occupation binds up to a size-limited capacity:

$$t_o(\mathbf{r}) = U(\mathbf{r})\,\frac{e_o(\mathbf{r})^{\beta_m}}{\sum_{o'} e_{o'}(\mathbf{r})^{\beta_m}}, \tag{8}$$

$$c_o = M_o/\theta_{\text{abs}}, \quad M_o = b_o^{\text{orig}} + r_o, \qquad \iota_o(\mathbf{r}) = t_o(\mathbf{r})\,\min\Big(1, \frac{c_o}{\int t_o\,d\mathbf{r}}\Big). \tag{9}$$

The claim $t_o$ distributes the unbound mass by sharpened match. The readiness $e_o$ is the shared attachment primitive of eq. (4), and the exponent $\beta_m > 1$ lets the best match dominate the claim at each location. The capacity $c_o$ scales with the occupation's task mass $M_o$, which grows endogenously with the reinstated mass $r_o$ it has already bound. What a saturated occupation cannot staff this step, the fraction $1 - \min(1, c_o/\int t_o)$, returns to $U$ and is offered at the next step to the next-best match. Large occupations therefore bind faster in absolute terms but not faster relative to their size, and the residue cascades down the match ranking.

The same primitive sets the static attachment capacity $C$ and allocates the unbound mass here; the static and dynamic layers of the implementation share it exactly. Its symmetry carries the economics. Being able to do a task and becoming its home are different questions, and penalising over-qualification alongside the deficit keeps an occupation far above a task's level from drawing the task in as its own work. Storck (2026) states the form and its one-sided limit ($\lambda_{\text{over}} = 0$, $\rho \to \infty$). What the dynamic layer adds is only the sharpening $\beta_m$ and the size-limited rate.

The form was chosen against an alternative that conflates the two roles. A binding rate proportional to size times match, $\iota_o \propto M_o\,e_o$, lets size *multiply* match. In that form the absolute absorption correlates with occupation size at +0.99 and with the unconstrained claim at +0.07 at the reference tempo, so size decides the destination and match governs only the relative growth. The match-allocated form with a size cap inverts this. Absorption tracks the claim at +0.73, and the size correlation falls to +0.13 at the reference tempo and to +0.04 in the gradual regime, with size demoted from setting *where* the work goes to *how fast* it binds, which is its proper role. The sharpness $\beta_m$ and the absorption timescale $\theta_{\text{abs}}$ are calibration handles rather than worldviews. Their economic content, whether absorptive capacity scales linearly or sub-linearly with size and how sharply the best match wins, is settled by a sweep, and the central result below is itself such a sweep. Their economic interpretation is direct. The sharpness $\beta_m$ sets how strongly the best match dominates the local claim: $\beta_m = 1$ shares the mass in proportion to fit, $\beta_m \to \infty$ assigns each location strictly to its best match; its calibration target is the observed concentration of new work across occupations. The absorption timescale $\theta_{\text{abs}}$ is the time an occupation needs to absorb its own task mass in new work, an onboarding capacity per unit size; its target is the observed pace at which occupations take on new task content. The runs fix $\beta_m = 3$ and sweep the tempo; Section 4.2 reports the result's sensitivity to $\beta_m$ over 1 to 8 and shows that the destination result is carried by $\theta_{\text{abs}}$ alone.

## 3.4 Limit allocations

The two roles that the binding law separates become exact in the two limits of the absorption tempo, and the destination result of Section 4 is the path between them.

**Proposition 1 (fast absorption)** *Fix the occupation set. As $\theta_{abs}/T_{shock} \to 0$ no capacity ever binds, and the final allocation of reinstated mass is*

$$R_o = \int w_o(\mathbf{r})\, s(\mathbf{r})\, d\mathbf{r}, \qquad w_o(\mathbf{r}) = \frac{e_o(\mathbf{r})^{\beta_m}}{\sum_{o'} e_{o'}(\mathbf{r})^{\beta_m}}, \tag{10}$$

*where $s$ is the cumulative surviving seed density. As $\beta_m \to \infty$ the shares concentrate on the best match and the allocation converges to strict best-match assignment.*

*Proof.* With slack capacities the absorbed fraction is one for every occupation at every step, so each step's seeding is bound in full at the claim shares of eq. (8). Absent births the readiness fields are time-invariant, so the shares $w_o$ are as well, and summing over steps gives $R_o = \int w_o\, s$. The second statement is the pointwise limit of the sharpened shares. □

The statement is exact at the level of the discrete law. As a self-consistency check of the discrete law against its own analytic limit, at $\theta_{\text{abs}} = 0.05$ the realised allocation matches $\int w_o\, s$ to machine precision (relative $L_1$ error below $10^{-15}$). The check verifies the implementation, not the model.

**Proposition 2 (slow absorption)** *As $\theta_{abs}/T_{shock} \to \infty$ the final allocation converges to the initial size shares, $R_o \to S\, M_o(0)/M_{tot}$, with $S$ the total surviving seed mass.*

*Proof.* Every claim share is strictly positive, since the readiness is an exponential, so each occupation's desired intake is positive wherever unbound mass remains, while the capacities $M_o/\theta_{\text{abs}}$ shrink without bound; for $\theta_{\text{abs}}$ large enough every capacity binds throughout the drain. While they bind, absorption follows $\dot{r}_o = M_o(t)/\theta_{\text{abs}}$ with $M_o(t) = M_o(0) + r_o(t)$, whose solution $r_o(t) = M_o(0)\,(e^{t/\theta_{\text{abs}}} - 1)$ is a common exponential factor times the initial mass: endogenous task-mass growth preserves the proportions. The phases in which some capacity is slack, the extreme tails of the seeding pulse and the end of the drain where the remaining unbound mass falls below the largest capacity, carry a mass share that vanishes as $\theta_{\text{abs}}$ grows. $\square$

While all capacities bind, the aggregate obeys $\dot{U}_{\text{tot}} = \dot{s}_{\text{tot}}(t) - M_{\text{tot}}/\theta_{\text{abs}}$, so after seeding ends the stock drains linearly and empties in $\theta_{\text{abs}}\, S/M_{\text{tot}}$ years. The bound is saturated only in the deep limit. The capacity utilisation, the mass absorbed per step over the aggregate bound $(\Delta t/\theta_{\text{abs}})\, M_{\text{tot}}$ read at the peak of the mismatch, rises from 0.11 at $\theta_{\text{abs}} = 2$ to 0.56 at 15 and 0.99 at 120. The shortfall is the spatial locality of binding: claims are proportional to $e_o^{\beta_m}$, so the mass on the seeding crescent is in practice drained by the occupations near it, and the aggregate task mass overstates the absorbing mass at intermediate tempos. This is why the mismatch hump of Section 4.1 outlives the naive aggregate prediction.

# 4 Results

We run a single technology field, calibrated to task-level AI exposure as in Storck (2026), through the dynamic economy. The field matures over $T_{\text{shock}} = 5$ years; we vary the redistribution tempo against this fixed shock, writing $\theta$ for the pooled setting $\theta_L = \theta_{\text{abs}} = \theta$. The two timescales are separated only in the decomposition check of Section 4.2. The economy parameters are the static model's, restated in Section 2; the dynamic layer adds the redistribution timescales $\theta_L$ and $\theta_{\text{abs}}$ (both 3 years at reference), the claim sharpness $\beta_m = 3$, the diffusion window $T_{\text{shock}} = 5$ years, and the step $\Delta t = 0.2$ years. Occupational birth is switched off in the runs of this section, so the occupation set is identical across regimes and the allocation comparison is well defined. In every run the population sum is conserved to numerical precision and the unbound stock drains below $10^{-8}$; both are asserted by the producing scripts.

## 4.1 The transitional mismatch

Figure 3 shows the shock and the mismatch it creates. The left panel is the logistic diffusion of (7), rising over the five-year window. The right panel is the unbound stock

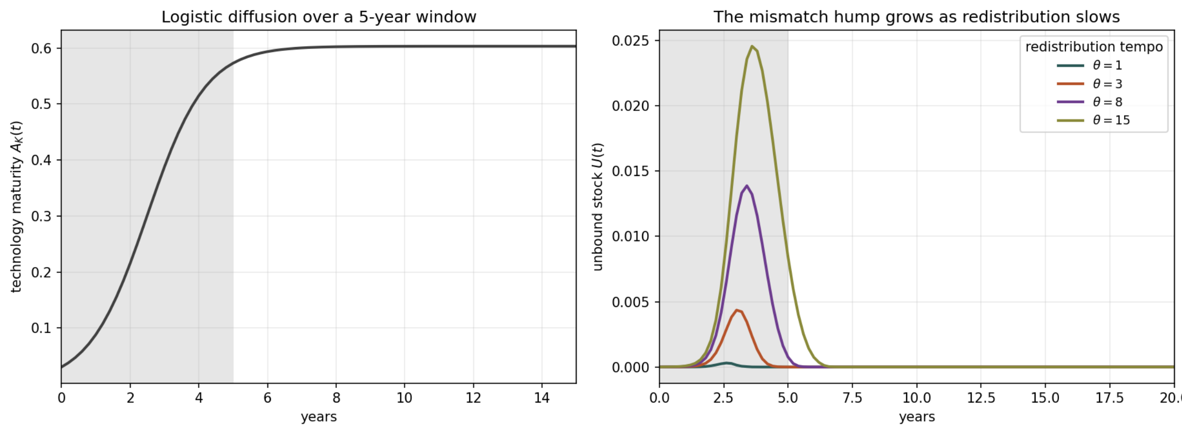


Figure 3: Calendar-time dynamics. Left: the technology matures as a logistic diffusion over a five-year window. Right: the unbound stock $U(t)$, the transitional mismatch between destruction and creation, for four redistribution tempos $\theta \in \{1, 3, 8, 15\}$ years against the fixed five-year shock. The mismatch hump grows and lingers as redistribution slows relative to the shock, but always drains to zero. The transitional cost is in the path rather than the endpoint.

$U(t)$ for four redistribution tempos, $\theta \in \{1, 3, 8, 15\}$ years. When redistribution is faster than the shock, the mismatch is barely visible, since the market binds the displaced work as fast as it is seeded. As redistribution slows, the mismatch hump grows by a factor of eighty from the fastest to the slowest tempo shown, and peaks later, moving from $t \approx 2.6$ to $t \approx 3.6$ years within the adoption window. In every case it drains to zero once diffusion saturates and binding catches up. The shape is that of the mismatch index Şahin et al. (2014) measure over the Great Recession, which likewise rises and drains with no structural shift. The driver here is task destruction and creation rather than the business cycle. There is no permanent mismatch in the model: all created work is eventually bound. The transitional cost lives in the height and persistence of the hump, not in the endpoint, and the static model cannot register it.

## 4.2 The destination depends on the tempo

The hump measures the cost of the transition; the same ratio also determines its outcome. Figure 4 shows where the reinstated work is bound in two regimes, run through the same technology and the same field. In the gradual regime ($\theta = 1$ year against the five-year shock), the best-matched occupations keep pace and absorb the work. The gainers are the high-skill specialists whose capabilities sit at the frontier of the field. In the congested regime ($\theta = 15$ years), the small best-matched occupations cannot bind fast enough. Their size-limited capacity $c_o = M_o/\theta_{\text{abs}}$ is small, the mass accumulates in $U$, and the large, lower-skilled occupations that can absorb it quickly take it instead. The same work is downgraded. Large low-skill service occupations have played the absorbing role before, and Autor and Dorn (2013) document their growth as routine work was automated away.

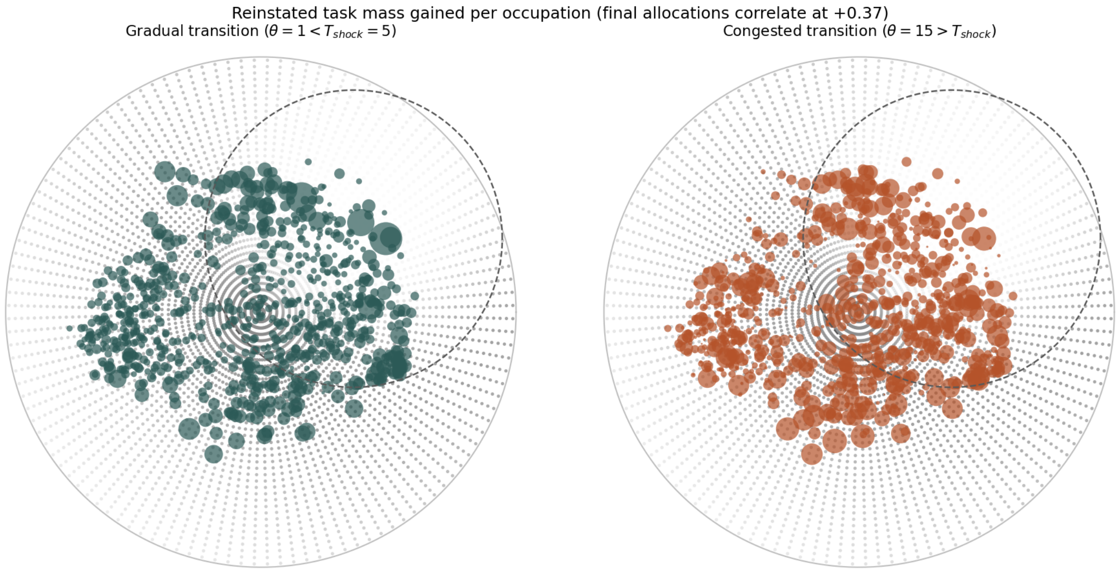


Figure 4: The destination of reinstated work depends on the tempo ratio. Both panels show the reinstated task mass gained by each occupation, over the survival-gated seeding field (grey) with the technology field dashed, for the same five-year shock. Left, a gradual transition: the best matches keep pace and high-skill specialists at the frontier absorb the work. Right, a congested transition: small best-matched occupations cannot keep up and the work is forced onto large, lower-skilled occupations. The two final allocations correlate at +0.37.

The leading gainers make the point concretely. In the gradual regime they are preventive-medicine physicians, physicists, and natural-sciences managers. In the congested regime they include fast-food and counter workers, waiters, and dishwashers.

The shift is not marginal. The final allocations of reinstated mass in the two regimes correlate at +0.37 (Pearson, unweighted across the 849 pre-existing occupations, +0.31 by rank). One technology and one field produce substantially different destinations, differing only in the tempo of the shock relative to the market's capacity to respond. Weighting by baseline employment raises the correlation to +0.68, because the largest occupations gain in both regimes. The divergence is concentrated in whether specialist best matches or large absorbers take the frontier work, which is the margin the result is about. This is the destination counterpart of the transitional hump. The fixed point sees the endpoint of a frictionless reallocation; the endpoint the economy actually reaches depends on how fast it was forced to move.

The mechanism reads off the per-occupation evidence. Define an occupation's unconstrained claim as the seed-weighted share the claim law (8) would allocate with no size cap, and its size as the pre-shock task mass that caps the rate. In the gradual regime absorption tracks the claim, with Pearson +0.83 and rank +0.92, and is uncorrelated with size (+0.04). The top claim quartile takes 49 percent of the reinstated mass. In the congested regime the claim decouples (+0.37, rank +0.21) and size takes over (rank

+0.95). The top size quartile takes 57 percent of the mass and the bottom quartile is squeezed from 22 to 4 percent. All rankings are in absolute mass; the relative-change measure is unbounded for small occupations and is not used. Rank agreement with size is not equality with global size shares: seeding is spatially local, so the congested allocation is an interior point between the limits, not the proportional allocation of Proposition 2.

Three robustness checks bound the claim. Sweeping the sharpness $\beta_m$ from 1 to 8 moves the cross-regime correlation from +0.26 to +0.61. The divergence of destinations survives the full range while its point size depends on the handle. Decomposing the pooled tempo into its two timescales shows which one binds. Slowing only absorption ($\theta_L = 1$, $\theta_{\text{abs}} = 15$) reproduces the congested allocation (correlation +0.39 with the gradual run), while slowing only mobility ($\theta_L = 15$, $\theta_{\text{abs}} = 1$) leaves the destination essentially unchanged (+1.00). The forced downgrading is a statement about how fast occupations can take on new work, not about how fast workers move.

The third check varies the survival gate itself, since the low-priced character of surviving seed is inherited from it. Both regimes are rerun with the gate off, and with a comparative-advantage gate, a human-productivity field $h(\mathbf{r}) = \exp[\lambda_h(1 - \hat{\phi}_K(\mathbf{r}))]$ with $\hat{\phi}_K$ the unit-amplitude technology shape, so labour is more productive where the machine is weak and capital operates where $s_K\phi_K > hR/\Pi$. The sweep takes $\lambda_h$ up to 2, and $\lambda_h = 0$ recovers the uniform-$R$ gate. The divergence is intact in every variant. The cross-regime correlation stays between +0.35 and +0.46, and the congested size rank correlation between +0.91 and +0.96. What the gate moves is the gainers' price level rather than the tempo dependence. Without the gate the capital-dominated core seeds, and the congested absorbers become large health and administrative occupations rather than food service. The comparative-advantage gate raises the mass-weighted gainer price with $\lambda_h$. The downgrading, congested gainers priced below gradual gainers, persists throughout.

## 4.3 The homotopy between the limits

The tempo result is the path between the two proved endpoints of Section 3.4. Figure 5 traces the final allocation across $\theta_{\text{abs}}$ from 0.5 to 120 years against the fixed five-year shock, with mobility held fast at $\theta_L = 1$, since the decomposition above shows the destination is carried by $\theta_{\text{abs}}$. Its correlation with the fast-limit allocation of Proposition 1 falls monotonically from +1.00 to +0.07, and its correlation with the size shares of Proposition 2 rises monotonically from +0.03 to +0.94 (+0.996 by rank) over the same sweep. The crossover, where the two associations are equal, sits between $\theta_{\text{abs}} = 10$ and 15, two to three shock windows. The simulation's remaining role is to locate the crossover, since the two endpoint allocations are derived.

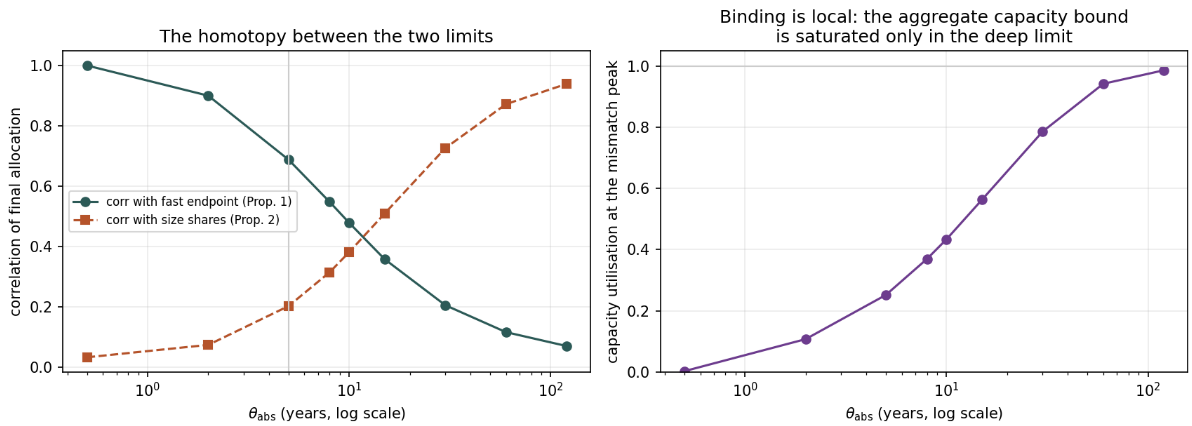


Figure 5: The destination as a homotopy between proved limits. Left: the correlation of the final allocation with the fast-limit claim allocation (Proposition 1) and with the slow-limit size shares (Proposition 2), across the absorption tempo at fixed mobility; the vertical line marks the shock window. Right: the capacity utilisation at the mismatch peak, the mass absorbed per step over the aggregate bound. The shortfall at intermediate tempos is the spatial locality of binding, and Proposition 2 is the regime where it reaches one.

# 5 The general-equilibrium price feedback

Both the static regime and the dynamic transition above hold the price field $\Pi(r)$ fixed. The shock relocates labour and re-sorts it, congestion adjusts the marginal value of crowded destinations, but the per-unit price of capability content does not re-solve. This is a short-run reading. The price field is an equilibrium object, derived as a capability assignment in Storck (2026). Its level $\Pi_0$ is the estimated field of eq. (2); holding it fixed discards part of the model's own content. Acemoglu and Restrepo (2026) make the point sharply. In their task model the base wage *adjusts* to automation through their propagation matrix $\Theta$, and that adjustment is the mechanism. Our fixed-price layers hold both the rent wedge and the base price fixed, so to "derive the prices" in their sense we must let the base price move. This section closes that loop.

**The closure: level from data, change from the assignment.** We keep the level from the estimated field, the empirical anchor, and take the *change* from the assignment, exactly as Acemoglu and Restrepo (2026) take the wage level as given and compute its response. Their wage law $w_g \propto (\Gamma_g/\ell_g)^{1/\lambda}$ ties the wage to tasks per worker, and its spatial form is

$$\Delta \ln \Pi(r) \;=\; \frac{1}{\sigma}\Big[\,\ln\big(1-a(r)\big) \;-\; \ln\frac{n_L(r)}{n_L^0(r)}\,\Big], \tag{11}$$

applied to the baseline, $\Pi(r) = \Pi_0(r)\, e^{\Delta \ln \Pi(r)}$. Automation is a *demand* shock to labour. Where capital takes the task ($a$ high), the task leaves the human set and the wage falls, and displaced labour crowding the remaining work ($n_L > n_L^0$) lowers it further. Both terms are stabilising. The sign is not innocuous. The opposite reading, automation as a *supply* shock that makes scarce human work dearer, inverts equation (11) and is destabilising, since a dearer wage opens the takeover gate. Run as a single pass at $\sigma = 3$, the scarcity reading collapses the labour share to 0.33 (capital chases the work it has just made dear), while the demand reading lifts it to 0.83. The fixed-price value 0.58 sits between. Labour shares in this section hold the allocation at its baseline, which isolates the price channel. The headline shares of Storck (2026) re-solve the allocation as well and are correspondingly higher, 0.626 at fixed prices and 0.78 at the fixed point below. The headline depends on the sign, which therefore must be chosen on economic grounds. Acemoglu and Restrepo (2026) choose ours, since automation dissipates rents and lowers wages.

**The static fixed point.** Iterating price and allocation to mutual consistency converges, and the convergence follows from a condition that can be evaluated at the calibration. Differentiating the gate (3) gives $\partial \ln(1-a)/\partial \ln \Pi = -a\,(R/\Pi)/\tau$. A lower price closes the gate, and the closing gate raises the price back. The slope of the map (11) is therefore bounded by

$$K \;\le\; \frac{1}{\sigma}\Big(\sup_{\mathbf{r}} \frac{a(\mathbf{r})\,R}{\Pi(\mathbf{r})\,\tau} \;+\; \varepsilon_n\Big), \tag{12}$$

where $\varepsilon_n$ is the crowding elasticity of the re-sorted density, and damped iteration with weight $d$ converges when the stabilising slope satisfies $K < 2/d - 1$. At $d = \frac{1}{2}$ the threshold is 3. Evaluated at the converged point, the gate term peaks at 1.64 and the spectral radius of the full map, crowding included, is 1.59 by power iteration. The condition holds with a factor-two margin, and the iteration converges in nine damped steps. The slope scales as $1/\sigma$ ($\sigma K = 4.77$ at $\sigma = 0.5$ against 4.78 at $\sigma = 3$), which places the failure boundary at $d = \frac{1}{2}$ near $\sigma = 1.6$. At $\sigma = 1$ the iteration indeed fails at $d = \frac{1}{2}$ and converges at $d = \frac{1}{4}$. The fixed point itself does not depend on the damping; only the path to it does.

The self-consistent state is an interior point rather than the one-pass extreme. The labour share settles at 0.73 (against 0.58 at fixed prices), the price level falls 26 percent, and displacement is cut by roughly 44 percent ($D_o$ from 0.28 to 0.16). The wage adjustment absorbs most of the shock. Human work becomes about a quarter cheaper, capital adopts far less, and the labour share stays high, but the harm now sits in the wage *level* rather than in employment, the rent- and wage-dissipation of Acemoglu and Restrepo (2026).

**The dynamic close, and statics meeting dynamics.** Wired into the transition, with the price responding to the current adoption and allocation at every step through equation (11) with a one-step lag, the feedback shapes the whole path. The wage level falls monotonically as adoption rises, from $-0.09$ early to a plateau at $-0.29$, and the displacement wave is markedly gentler. The unbound-mass peak falls by about 63 percent (from 0.44 to 0.16 percent of the workforce) at the same timing, and the reinstated stock ends lower, because the falling wage deters the automation that would have produced the mismatch. The transition lands where the static fixed point says it should. The dynamic end-state labour share (0.739) matches the static fixed point (0.73); the price levels ($-0.285$ against $-0.26$) differ by about ten per cent of the value. Two independent constructions, static fixed-point iteration and dynamic time-stepping, land within this gap of the same equilibrium, so the price feedback is a property of the model rather than of either method, and the static and dynamic accounts cohere.

**What this does and does not establish.** The feedback is first-order. At fixed prices both layers overstate the displacement and understate the labour share, and they overstate the violence of the transition. The stabilising wage adjustment is the shock absorber, and it operates throughout the transition. The quantitative point estimate depends on the assignment elasticity $\sigma$ and on the reduced-form closure, since equation (11) is the AR wage law applied region-wise rather than a full re-solve of the assignment programme of Storck (2026). The AI field remains one illustrative instance of a technology field rather than a measurement. Across $\sigma$ from 5 to 0.5 the fixed-point lift of the labour share runs from $+0.11$ to a peak of $+0.20$ at $\sigma = 1$, and the damping of the mismatch hump rises from 46 to 89 per cent. We therefore read the result as establishing the *direction and order of magnitude* of the price feedback, and the mutual consistency of the static and dynamic closes, rather than a calibrated forecast of the labour share.

# 6 Discussion

The result is that the tempo of a technology shock, relative to the speed at which the labour market reallocates, determines where the reinstated work ends up. A gradual diffusion lets work find the occupations whose capabilities match it; a shock that outruns adjustment forces a downgrading that persists in the final allocation. The same field and the same seeded reinstatement yield different allocations of work depending on the pace. The task-based equilibrium framework is not built to make this statement. Its rest point is the destination of a frictionless economy, and the friction here is time itself.

Two qualifications bound the claim, and both are inherited rather than introduced. The first concerns the survival gate. The static model places surviving reinstatement on the low-priced side of the boundary, because a seeded task is retained by labour where labour is the cheaper producer at a *uniform* cost $R$. In Acemoglu and Restrepo (2019) reinstatement is instead new *complex* work in which labour has a comparative advantage, which requires the survival condition to turn on relative rather than absolute productivity. Section 4.2 runs exactly this variant, alongside the gate removed altogether, and the gate turns out to govern the *character* of the destination rather than its tempo dependence. The divergence and the size mechanism survive every variant while the gainers' price level moves with the gate. What remains conditional on the survival-gate economics is therefore the high- versus low-skill reading of the congested absorbers, food service under the absolute gate and large health and administrative occupations without it, rather than the result that a fast shock forces the work onto whoever is large enough to take it.

The second qualification is the binding calibration. The allocation depends on the sharpness $\beta_m$ and the absorption timescale $\theta_{\text{abs}}$, and Section 4.2 reports both sensitivities. The divergence of destinations survives the sharpness sweep with a point size that moves with the handle, and the absorption timescale alone carries it. The remaining quantitative claim is the regime boundary, how slow absorption must be relative to the shock for downgrading to set in, which the tempo sweep exhibits rather than derives.

Several limitations remain. The realisation is illustrative, inherits the calibration and discipline of Storck (2026), and supports no causal claim. The unbound stock drains to zero, so the model has no permanent mismatch and no involuntary non-employment. The transitional cost is the hump, and a model with task decay or irreversible binding would carry a permanent imprint of the shock speed that this one does not. The imprint is empirically real and uneven, since Edin et al. (2023) find moderate mean earnings losses from occupational decline, concentrated on low earners. That extension is left for later work. New tasks enter as added weight on existing locations rather than as new points on the disk. And the relative-change measure of who gains is unbounded for small occupations, so the destination results should be read in absolute and spatial terms.

The downgrading of the fast regime has an empirical silhouette. Beaudry, Green, and Sand (2016) document a cascade after 2000. As demand for cognitive tasks fell, high-educated workers moved down the occupational ladder and pushed the less educated further down and out. The direction is reversed here. In their account workers descend the ladder because demand for their tasks has fallen; in the fast regime the work itself descends the match ranking because the occupations that fit it best are too small to absorb it in time. The same outward pattern, work performed below the skill level that matches it, arises from different mechanisms, one demand-side and one absorption-side, and the cascading image is theirs.

# 7 Conclusion

Turning the spatial task-based model from an equilibrium into laws of motion requires one new object, a stock of unbound task mass, and yields two things the equilibrium cannot. The first is the transitional mismatch $U(t)$, the gap between destruction and creation as a transient that grows with the tempo of the shock and drains as the market catches up. The second, and the central result, is that the tempo of the shock relative to the speed of redistribution determines the destination of the reinstated work: a gradual shock matches it to high-skill specialists, a fast shock forces it onto large lower-skilled occupations. Both endpoints are proved as limits of the binding law, and the realised destination moves monotonically between them with the tempo. The same technology yields a different distribution of work depending only on how fast it arrives. The cost and the outcome of automation are, in this sense, properties of the path and not only of the endpoint.

# A The inherited economy

The dynamics consume the static economy across one interface, and this appendix restates everything that crosses it, so that every parameter of the model traces to a stated source within this paper. The construction and validation of the geometry remain with Storck and Andersson (2026); the model of technological change and its calibration with Storck (2026). Table values reproduced here are adapted from those papers. Table 2 consolidates the parameters; the sections below state where each number comes from.

## Geometry

The task space is built from the 17,606 task statements of O*NET, each encoded by a text-embedding model as a vector in a high-dimensional semantic space. The embedding matrix is centred and projected on its first two principal components, and the scores are read in polar coordinates: the angle $\xi$ is a domain orientation, the radius $\chi \in [0, 1]$ a normalised specificity. The orientation of the map is frozen against a fixed reference, so runs, encoders, and database releases are comparable. Occupations enter as importance-weighted bundles $b_o(\mathbf{r})$ of their task points with centroid $\mu_o$; 849 occupations with observed employment enter the dynamic economy, weighted by their OEWS employment shares $L_0$. The choice of encoder, the recovery of the expert-curated job-family classification, and the external validity of the coordinates are established in Storck and Andersson (2026) and not reproduced here.

## The price field

The price of skill is the reduced form of eq. (1), estimated once on the occupational wage cross-section (785 occupations with observed mean wages) and held fixed under the shock. The coefficients and their HC3 standard errors are those of eq. (2), with $R^2 = 0.523$. Two structural checks discipline the form. The isotropic depth term is indistinguishable from zero ($m_3 = -0.065$, $p = 0.47$): depth pays only through its interaction with direction. A second-harmonic test balanced with level harmonics does not reject the single first harmonic of the depth return ($p = 0.44$); the remaining higher-order structure is a level effect, not additional depth pricing. The step from centroid wages to task-location prices is validated by bundle integration: pricing an occupation as $\int \Pi \, b_o$ reproduces observed

Table 1: Capability plane coefficients, $q_k(\mathbf{r}) = \bar{v}_k + \chi\,(u_{1,k}\cos\xi + u_{2,k}\sin\xi)$. $R^2$ is the plane fit; $\Delta R^2$ the gain from level harmonics and isotropic depth. Only $S1$ and $S2$ are priced. Adapted from Storck (2026).

| Cluster | $\bar{v}_k$ | $u_{1,k}$ | $u_{2,k}$ | Pole | $R^2$ | $\Delta R^2$ |
|---|---|---|---|---|---|---|
| S1 | 2.68 | 1.09 | 1.42 | 52° | 0.74 | 0.001 |
| S2 | 1.37 | −2.22 | 1.23 | 151° | 0.69 | 0.008 |
| A1 | 3.20 | 0.95 | 1.31 | 54° | 0.75 | 0.000 |
| A2 | 1.57 | −1.54 | −0.24 | 189° | 0.57 | 0.009 |

wages about as well as evaluating the field at the centroid ($R^2$ 0.51 against 0.52), which licenses operating the field at task locations.

## Readiness

The readiness primitive of eq. (4) needs the requirement fields $q_k$, the weights $v_k$, and three scalars. Each capability cluster's requirement field is a plane over the disk, $q_k(\mathbf{r}) = \bar{v}_k + \chi\,(u_{1,k}\cos\xi + u_{2,k}\sin\xi)$; Table 1 restates the coefficients. The gain from adding level harmonics and isotropic depth to the planes is at most $\Delta R^2 = 0.009$, so the plane is not a restrictive form. The weights $v_k$ are wage returns from a replication of the mediation regression of Storck and Andersson (2026) on this sample: $v_{S1} = 0.328$ (0.065), $v_{S2} = 0.049$ (0.021). The ability clusters carry negative or negligible estimates and are excluded by the sign restriction $v_k \geq 0$, so the priced set is $K = \{S1, S2\}$. The acquisition scale $\ell = 0.133$ is one within-direction standard deviation of the priced-capability index, the deficit at which readiness falls to $e^{-1}$. The locality $\rho = 0.5$ and the over-qualification weight $\lambda_{\text{over}} = 1$ are set, not estimated.

## The technology field

A technology is the Gaussian field of Section 2 with centre $\mathbf{p}_K$, reach $z_K$, and mature amplitude $\bar{A}_K$. The field run in Section 4 is calibrated to task-level AI exposure: each task statement carries the exposure score $\beta_E = E1 + \frac{1}{2}E2$ of Eloundou et al. (2024), the scores are smoothed to a between-cell exposure surface on the disk, and the Gaussian is fitted to that surface by least squares. The fit places the centre at $(\xi_K, \chi_K) = (38.4°, 0.463)$ with reach $z_K = 0.583$ and mature amplitude $\bar{A}_K = 0.603$, and explains $R^2 = 0.82$ of the smoothable between-cell surface (0.25 of raw task-level exposure). The projection is one instance of a technology field, an illustration rather than a measurement; nothing in the dynamics depends on the field being AI. The scale $s_K$ is normalised to one, and the maturation $A_K(t)$ rises from zero to $\bar{A}_K$ along the logistic of eq. (7).

Table 2: Parameter provenance. The wage wedge $\eta_g$ and the texture ledger of Storck (2026) do not cross the interface and are not used by the dynamics.

| Parameter | Value | Provenance |
|---|---|---|
| $m_0, \ldots, m_5$ | eq. (2) | estimated: wage cross-section, $N = 785$, HC3, $R^2 = 0.523$ |
| $v_{S1},\ v_{S2}$ | 0.328, 0.049 | estimated: mediation-regression replication, s.e. 0.065, 0.021; $v_k \geq 0$ excludes A1, A2 |
| $\bar{v}_k, u_{1,k}, u_{2,k}$ | Table 1 | estimated: capability plane fits |
| $\ell$ | 0.133 | rule: one within-direction SD of the priced-capability index gives readiness $e^{-1}$ |
| $\rho,\ \lambda_{\text{over}}$ | 0.5, 1 | set: attachment locality and over-qualification weight |
| $R$ | 18 | calibrated: the gate binds in the calibrated core, capital cost $\approx 30$ at the centre against $\Pi \approx 47$ |
| $\tau$ | 0.08 | set: width of the adoption margin, eq. (3) |
| $\beta$ | 0.5 | normalised: congestion benchmark |
| $\gamma$ | 0.5 | set: seeded fraction of displaced mass; swept in the static companion |
| $\nu$ | 11.8 | rule: one SD of baseline occupation value per logit unit, evaluated at $A_K = 0$ |
| $c_m$ | 23.0 | rule: the median move costs one $\nu$ |
| $\xi_K,\ \chi_K$ | 38.4°, 0.463 | calibrated: least-squares fit to the exposure surface |
| $z_K$ | 0.583 | calibrated: same fit |
| $\bar{A}_K$ | 0.603 | calibrated: same fit |
| $s_K$ | 1 | normalised |
| $\theta_L,\ \theta_{\text{abs}}$ | 3 yr | reference; swept in Section 4 |
| $\beta_m$ | 3 | reference claim sharpness; swept 1–8 |
| $T_{\text{shock}}$ | 5 yr | normalisation of the time axis |
| $\Delta t$ | 0.2 yr | numerical step |
| $\sigma$ | 3 | reference assignment elasticity; swept 0.5–5 |

## Parameter provenance

Table 2 consolidates the economy. Estimated parameters carry their regression source and standard error; calibrated parameters their target; rule-derived parameters their rule; the rest are normalised, set, or swept, as stated.

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